PATENT-STYLE TECHNICAL DISCLOSURE

Adaptive Hydrothermal Metal Recovery Reactor

with Integrated Reactive Transport and Spatially Controlled Selective Deposition

This draft is structured in the figure-oriented style of the supplied patent-format document. It describes a process and apparatus concept; it is not a grant of patent rights or a conclusion that the claims are novel or patentable.

1. TITLE

Adaptive Hydrothermal Metal Recovery Reactor with Continuous Reactive Transport and Spatially Controlled Selective Metal Deposition

2. FIELD OF THE INVENTION

The invention relates to hydrothermal and hydrometallurgical recovery of metals from mineral-bearing materials. More particularly, it relates to a continuous reactor pathway in which metal-bearing hydrothermal fluid is transported through multiple controlled reaction zones that alter physicochemical conditions and induce metal precipitation at selected locations.

3. BACKGROUND

The supplied reference describes hydrothermal buffering as a natural geochemical process in which hot, mineral-rich fluid interacts with rock, allowing metals to dissolve, remain in solution during transport, and subsequently precipitate when conditions change. It identifies temperature, pH, pressure, rock type, fluid composition, and redox conditions as important factors.

The reference presents five conceptual stages: heating and fluid generation, rock interaction (buffering), metal transport in solution, physical/chemical changes, and metal deposition. The present proposal does not claim that these natural phenomena are themselves new. Instead, it proposes an engineered reactor architecture that integrates transport and controlled deposition into a continuous pathway.

Prior patent literature also shows that hydrothermal extraction and precipitation can be controlled through temperature, pH, fluid chemistry, multiple reaction vessels, and precipitation zones. Accordingly, the particular reactor configuration and control strategy must be distinguished from such prior art to support any future patent claim.

4. SUMMARY OF THE INVENTION

The invention provides a continuous hydrothermal metal-recovery system comprising a metal-generation region, a reactive transport pathway, multiple deposition zones, sensors, a controller, and controllable actuators.

A heated fluid contacts mineral-bearing material and becomes enriched in dissolved metals. Instead of transporting the fluid to a separate batch precipitation operation, the metal-rich fluid enters a continuous pathway containing spatially separated reaction zones. Each zone is configured to alter one or more parameters such as temperature, pH, pressure, fluid composition, or redox condition. These changes reduce the solubility of selected metals and cause precipitation at controlled locations.

In an optional embodiment, sensors measure process conditions and a controller adjusts heating, cooling, fluid addition, reagent addition, flow, or pressure so that precipitation occurs within a desired zone.

5. FIGURE 1: SYSTEM VIEW

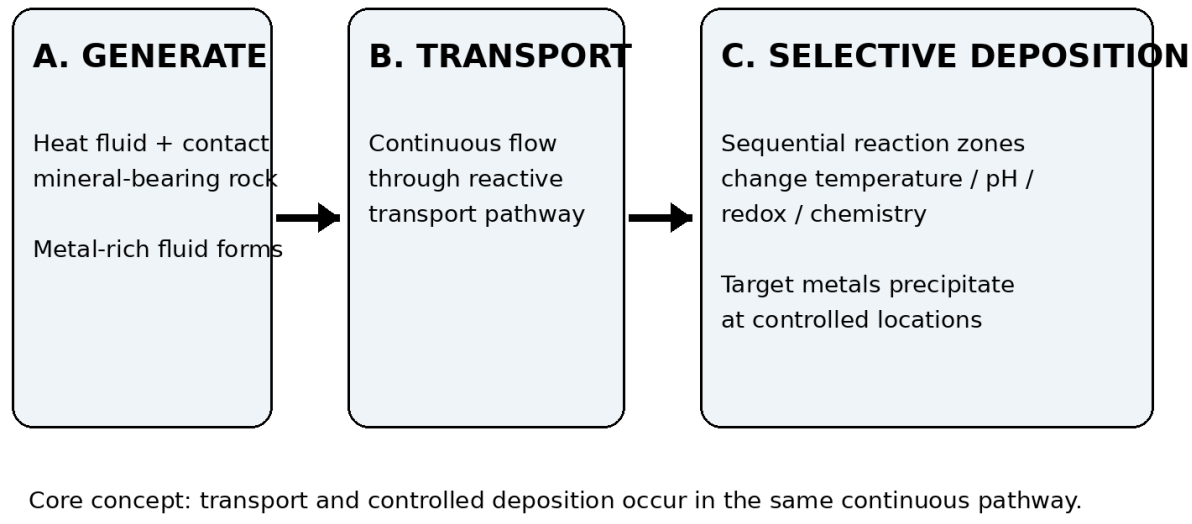


Figure 1 illustrates the three principal functional regions of the proposed system.

6. FIGURE 2: REACTIVE DEPOSITION CHANNEL

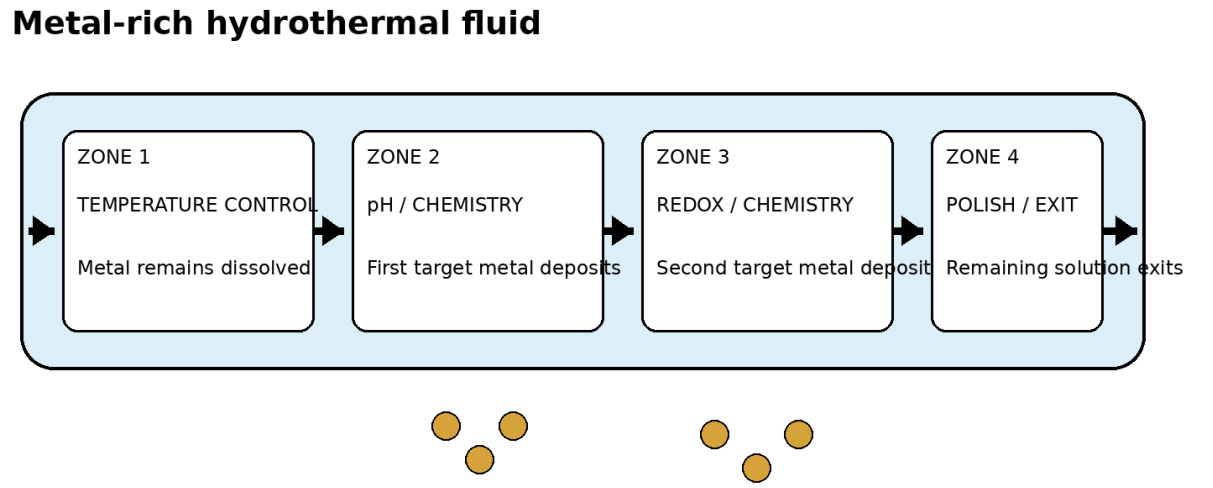


Figure 2 illustrates a continuous channel containing spatially separated reaction zones.

7. FIGURE 3: SENSOR FEEDBACK CONTROL

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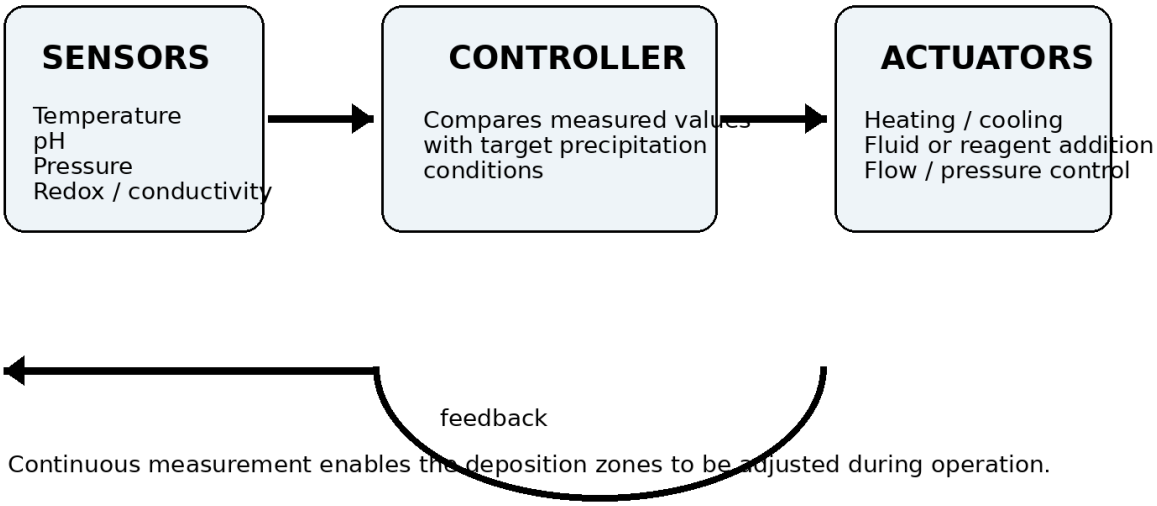


Figure 3 illustrates optional closed-loop control of the reactive deposition zones.

8. DETAILED DESCRIPTION

8.1 Metal generation. A carrier fluid is heated to a selected hydrothermal operating condition and contacted with mineral-bearing rock, ore, tailings, or another metal-containing solid. The interaction promotes dissolution or mobilization of one or more target metals into the fluid.

8.2 Continuous transport. The metal-bearing fluid is maintained in flow through a conduit, fracture network, porous medium, packed bed, tubular reactor, or other fluid pathway. The pathway is designed to preserve sufficient metal solubility until the fluid reaches a designated reaction zone.

8.3 Reactive deposition zones. The pathway includes two or more zones having different operating conditions. A first zone may change temperature, pH, pressure, chemical composition, or redox state. A subsequent zone may apply a different condition. The purpose is to create controlled precipitation locations rather than allowing uncontrolled deposition throughout the transport pathway.

8.4 Selective recovery. Because different metals have different solubility responses to physicochemical conditions, the zones may be configured so that a selected metal or metal group precipitates preferentially in one zone while other metals remain dissolved for subsequent transport. The exact operating conditions are to be determined experimentally for the selected ore/fluid chemistry.

8.5 Feedback control. Sensors can measure temperature, pH, pressure, conductivity, oxidation-reduction potential, flow rate, or another process variable. A controller compares measured values with target operating ranges and commands actuators to modify heating, cooling, flow, pressure, or addition of fluid/reagent. This allows the deposition zone to be stabilized or shifted during operation.

8.6 Collection. Precipitated material can be collected from removable liners, deposition surfaces, settling sections, filters, separators, or other collection structures associated with the reactive zones.

9. CORE TECHNICAL ADVANTAGE

The principal proposed improvement is the integration of metal transport and controlled precipitation within the same continuous reactor pathway. Rather than treating transport and deposition as disconnected operations, the system creates spatially controlled physicochemical transitions while the metal-bearing fluid remains in continuous motion.

Potential benefits include reduced process interruptions, reduced separate precipitation equipment, controlled deposition locations, easier collection of precipitates, and the possibility of sequential recovery of different metals. These benefits are engineering objectives and require experimental validation.

10. PRELIMINARY CLAIMS

1. A hydrothermal metal-recovery system comprising: a fluid-heating region configured to generate a heated fluid; a metal-contact region configured to contact the heated fluid with mineral-bearing material to form a metal-bearing fluid; a continuous reactive transport pathway receiving the metal-bearing fluid; and a plurality of spatially separated deposition zones configured to alter at least one physicochemical condition of the metal-bearing fluid and thereby cause precipitation of at least one dissolved metal at a controlled location within the pathway.

2. The system of claim 1, wherein the physicochemical condition comprises temperature, pH, pressure, fluid composition, oxidation-reduction condition, or a combination thereof.

3. The system of claim 1, wherein at least two deposition zones operate at different physicochemical conditions.

4. The system of claim 1, wherein a first deposition zone is configured to precipitate a first metal or metal group while a second deposition zone is configured to precipitate a second metal or metal group.

5. The system of claim 1, further comprising one or more sensors configured to measure temperature, pH, pressure, flow, conductivity, oxidation-reduction potential, or another process variable.

6. The system of claim 5, further comprising a controller configured to receive sensor measurements and control at least one actuator to modify heating, cooling, fluid addition, reagent addition, flow, or pressure.

7. A method of recovering metals comprising: generating a metal-bearing hydrothermal fluid by heating a carrier fluid and contacting the carrier fluid with mineral-bearing material; continuously transporting the metal-bearing fluid through a plurality of reaction zones; changing at least one physicochemical condition between reaction zones; and collecting metal precipitate formed within at least one reaction zone.

8. The method of claim 7, wherein the physicochemical condition is changed without interrupting continuous transport of the metal-bearing fluid.

9. The method of claim 7, wherein sensor feedback is used to maintain precipitation within a selected reaction zone.

10. The method of claim 7, wherein the mineral-bearing material contains one or more metals selected from copper, zinc, lead, nickel, cobalt, silver, gold, molybdenum, tungsten, tin, uranium, and rare earth elements.

11. NOVELTY / PRIOR-ART POSITION

This draft should not be represented as proving a new invention. A preliminary web search located prior patents describing hydrothermal mineral extraction with precipitation induced by temperature or pH changes, as well as hydrometallurgical systems using multiple reaction stages and selective precipitation. Therefore, the potentially distinguishing subject matter should focus on the specific continuous reactor architecture, spatially controlled deposition zones, and any demonstrated feedback-control mechanism or measurable performance improvement.

Before filing, a patent professional should conduct a formal prior-art search and revise the claims around features that are demonstrably novel and non-obvious.

12. ORIGINALITY STATEMENT

The proposed combination was developed for this technical disclosure as an engineered improvement concept. It should not be described as independently proven novel until a formal prior-art search and technical validation have been completed.

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13. INVENTOR / APPLICANT DETAILS

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Institution / Organization: _____

Date: _____

Signature: _____